SUBJECT VOCABULARY

minijack a plug and socket widely used for analogue audio signals in portable devices
HDMI (High-Definition Multimedia Interface) used to transmit video and audio data



▲ Figure 4.3 Device-to-device communication

▼ Table 4.1 Examples of device-to-device communication

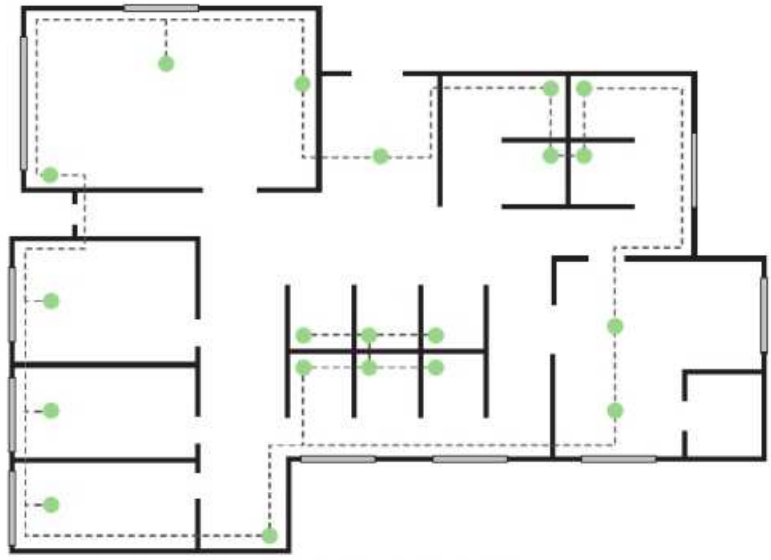
DEVICE 1	DEVICE 2	CONNECTION BETWEEN DEVICES 1 AND 2	USE
Temperature sensor	Air conditioner	Wired	To turn on the air conditioning when the temperature is too high
Smartphone	Headphones	**Minijack**	To play music from the smartphone on the headphones
Laptop	External hard drive	USB	To transfer files
Camcorder	Monitor	**HDMI**	To operate as a security camera
Games controller	Games console	Bluetooth	To control a game

NETWORK COMMUNICATION

When two or more computers are connected, a network is created. There are four major types of network.

LOCAL AREA NETWORK (LAN)

A LAN is a network that connects digital devices that are in a small geographical area, like a building or group of buildings that are close to each other.



▲ Figure 4.4 LANs are found in homes, schools and office buildings

WIDE AREA NETWORK (WAN)

A WAN is a network that is spread over a large geographical area. WANs are often used to connect different buildings owned by national and international businesses, law enforcement agencies, health and education organisations and government departments. Some organisations launch their own satellites to provide connectivity for their own global WANs.

WANs often use connectivity provided by a third-party telecommunications company, often linking LANs together through the internet. Because of their wider reach, WANs often have slower transfer speeds than LANs.

DID YOU KNOW?
The internet is the largest example of a WAN.



▲ **Figure 4.5** The internet is the largest example of a WAN

PERSONAL AREA NETWORK (PAN)

A PAN is a group of connected devices that are all near an individual user. For example, a user could connect their smartwatch to their smartphone, which is connected to their laptop and home cinema speaker system. Devices in a PAN can either be connected to each other directly or connected through access points (see page 84 for more information about access points).

When a PAN only uses wireless connectivity, it can also be referred to as a WPAN (**W**ireless **P**ersonal **A**rea **N**etwork). However, the general term PAN is more commonly used to refer to all types of PAN.

TETHERING

SUBJECT VOCABULARY

tethering connecting a host device that uses a mobile broadband connection with other devices so that they can use the host's broadband connection

Tethering is the process of connecting a host device, such as a smartphone or a tablet device, that uses a mobile broadband connection with one or more other devices. This enables the other device or devices to share the host device's broadband connection.



▲ **Figure 4.6** Devices can be tethered using wired or wireless connectivity

SUBJECT VOCABULARY

service agreement contract

Mobile phone network providers can enable or disable tethering as part of the **service agreement**. Some network providers charge more for this feature to be enabled.

5 NETWORKS

A network is created when two or more computers are connected together. Using a network, a computer can communicate with others and share resources such as hardware, software and data. It does this by sending data in a packet, which is a unit of data packaged together so that it can travel across a network.

The first network was created in 1960 by a commercial airline when it connected two mainframe computers together in order to speed up its reservation system. In 1962, researchers at the Advanced Research Projects Agency (ARPA) developed a project called the 'Intergalactic Computer Network'. In 1969, computers at four universities in the USA were connected, which created the ARPANET. This is considered to have been the predecessor of the internet.



LEARNING OBJECTIVES

- Know about network operating systems and how devices are identified on a network: device name, internet protocol (IP) and machine address code (MAC)
- Understand the function of components of wired and wireless systems: cable, wireless access point, router, gateway, booster, server
- Know the role of these for connecting to and using the internet:
  - web browser
  - ISP
  - search engine
  - filter software
- Know about peer-to-peer and client-server networks
- Know about the role of servers in a client-server network
- Understand the benefits of using local area network:
  - shared peripherals
  - shared data
  - flexible access
  - media streaming
  - communication
  - shared access to the internet
- Understand the benefits of using a client-server network:
  - control of user access rights
  - centralised administration
  - centralised backup
  - shared software
  - shared storage and file access
  - roaming profiles (hotdesk)
- Know about and understand the use of logins and passwords, firewalls, WEP/WPA, encryption, VPN, file access rights, transaction logs and backups
- Be able to select suitable methods of securing data for a particular context

SUBJECT VOCABULARY

protocols rules that allow the exchange and transmission of data between devices

REQUIREMENTS FOR CONNECTING TO NETWORKS

In order to connect to a network, computers need to fulfil certain requirements so that they all operate using standard **protocols**.

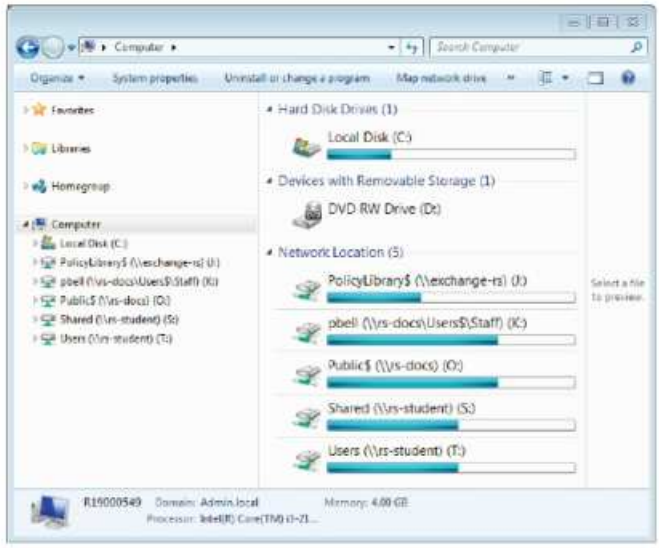
NETWORK OPERATING SYSTEMS

GENERAL VOCABULARY

stand-alone capable of functioning on its own, such as a stand-alone computer working on its own without being part of a network

A network operating system allows a computer to communicate on a network. It provides additional functionality to a **stand-alone** operating system, including:

- passing usernames and passwords to a server for checking when a user logs in
- separating user accounts and ensuring that users cannot access each other's files
- providing access to network storage and shared resources such as networked printers.



▲ **Figure 5.1** Network operating systems provide access to shared storage that is available on the network

SKILLS CRITICAL THINKING EXECUTIVE FUNCTION

ACTIVITY

▼ COMPARING OPERATING SYSTEMS

Create a table to compare the features of a network operating system to a stand-alone operating system.

HOW DEVICES ARE IDENTIFIED ON A NETWORK

There are three methods used to identify devices on a network:

- Internet Protocol (IP)
- MAC address
- device name.

SUBJECT VOCABULARY

IP address a unique address that networked devices use to send data to each other

INTERNET PROTOCOL (IP)

An **IP address** is a unique address that networked devices use to send data to each other. Each piece of data that is sent across a network carries the IP address of the destination, so that each device in the network knows where to send it.

SUBJECT VOCABULARY

hexadecimal a base-16 number system that uses the numbers 0–9 and the letters A–F
network administrator a person who manages an organisation's network

DID YOU KNOW?

IPv4 can store over 4 billion addresses. It was developed because of the huge growth in the number of devices that were being connected.

GENERAL VOCABULARY

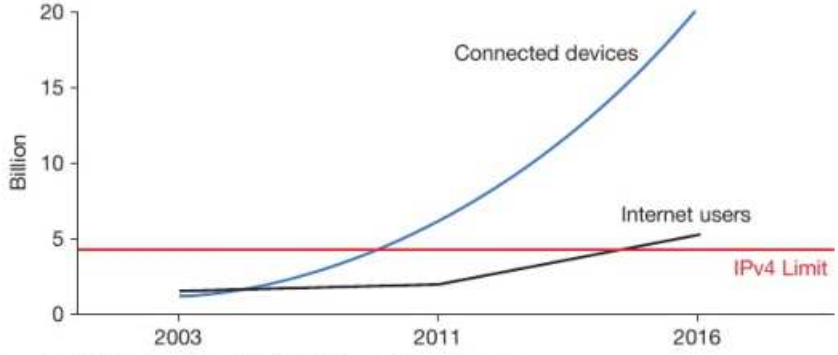
dynamically in a way that is open to change
Dynamic Host Configuration Protocol (DHCP) server a networked computer that automatically assigns an IP address to other computers when they join the network

SKILLS
INTELLECTUAL INTEREST AND CURIOSITY
CRITICAL THINKING
REASONING
COMMUNICATION
INTERPERSONAL SKILLS

IP addresses are made up groups of numbers. There are two main versions of IP in use.

- **IPv4** uses four groups of up to three numbers separated by full stops (for example, 192.168.1.1).
- **IPv6** uses eight groups of four **hexadecimal** numbers separated by colons (for example, 2001:0db8:0000:0042:0000:8a2e:0370:7334).

IP addresses can either be assigned by a **network administrator** or allocated **dynamically** by a server running **Dynamic Host Configuration Protocol** (DHCP).

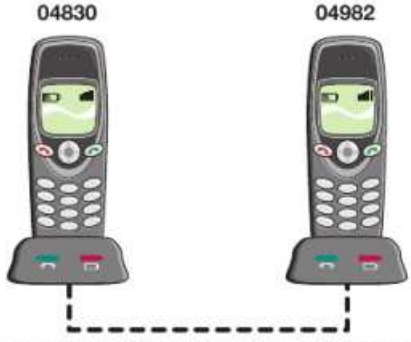


▲ **Figure 5.2** IPv4 can hold 4,294,967,296 possible addresses

ACTIVITY

▼ IP ADDRESSES

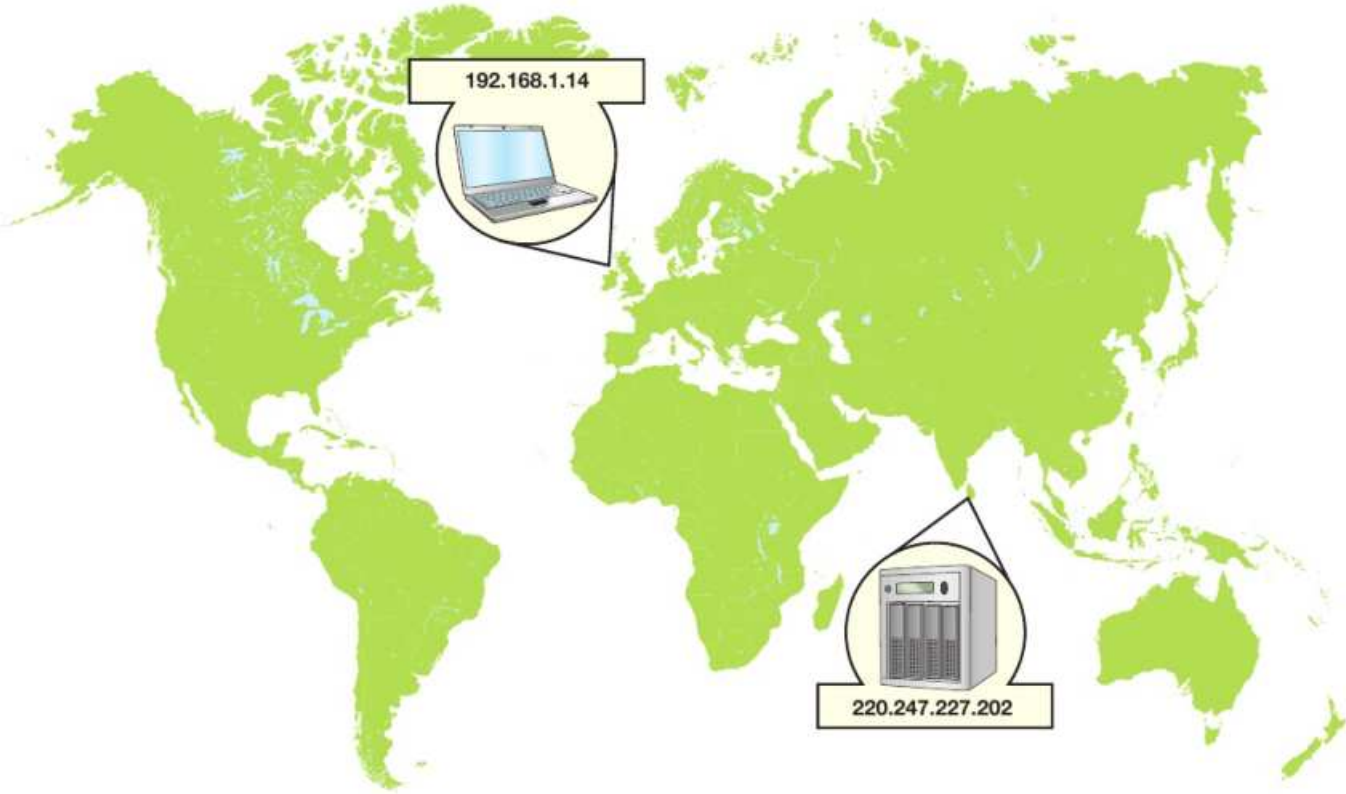
1 Research the maximum number of addresses that can be provided by IPv6.
2 Discuss why the number of connected devices is growing at a faster rate than the number of internet users.



▲ **Figure 5.3** When you make a telephone call, you dial a number that identifies the destination telephone



▲ **Figure 5.4** When you post a letter, you write the address on the letter so that every part of the postal system knows where to send it



▲ **Figure 5.5** When sending data, devices include the destination address of the data, just like in the telephone or postal system

GENERAL VOCABULARY

universally unique the only one of its kind in the world

SUBJECT VOCABULARY

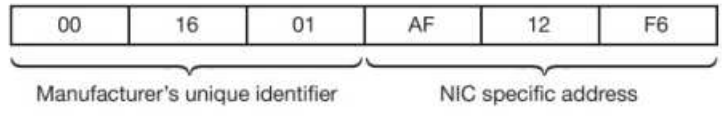
identifier a group of letters, numbers, or symbols that a computer has been programmed to recognise and uses to process information

MAC ADDRESS

Unlike IP addresses, which can be dynamically allocated by users or servers, media access control (MAC) addresses are **universally unique identifiers** given to the network interface card (NIC). MAC addresses are used in devices connected to local area networks (LANs) using Ethernet, Bluetooth or Wi-Fi.

MAC addresses are assigned by the NIC manufacturer and are generally considered to be fixed addresses.

A MAC address is usually made up of six pairs of characters. The first three pairs identify the manufacturer and the remaining pairs are assigned by the manufacturer to uniquely identify the device. Figure 5.6 is an example of a MAC address.



▲ **Figure 5.6** A MAC address

SKILLS SELF-DIRECTION PROBLEM SOLVING

ACTIVITY

▼ USING MAC ADDRESSES

Using the internet and other resources, find out which manufacturer made the device with the MAC address in Figure 5.6.

SUBJECT VOCABULARY

hotspot a place in a public building where there is a computer system with an access point, which allows people in the building with a wireless computer or Bluetooth mobile phone to connect to a service such as the internet

▼ Table 5.1 Uses of MAC addresses

USE	EXAMPLE
Restricting or allowing access to a network	MAC address filtering checks the MAC address of devices attempting to gain access to a network and only grants access to devices with specified MAC addresses
Identifying a device on a network	Some Wi-Fi hotspots only provide free access for a certain length of time, and they identify a device using its MAC address in order to work out whether it is trying to access the hotspot for longer than the permitted time
Tracking a device	Some companies and organisations track devices (and therefore their users) by checking which wireless access points have been accessed by specific MAC address
Assigning 'static' or 'fixed' IP addresses	Each time a device connects to a network, it is identified by a DHCP server (usually using its MAC) and given the same IP address as before

SKILLS INTELLECTUAL INTEREST AND CURIOSITY, CRITICAL THINKING, REASONING, COMMUNICATION, INTERPERSONAL SKILLS, PERSONAL AND SOCIAL RESPONSIBILITY

ACTIVITY

▼ SPOOFING

MAC addresses can be changed through a process called spoofing, which means using the MAC address of another device. Discuss how spoofing would affect each of the uses listed in Table 5.1.

DEVICE NAME

A device name is a descriptive name that helps users to identify computers on a network. Device names are not used by computers to communicate with each other as they are not always unique. This means that they could cause conflicts if data was sent to more than one device with the same name for processing.

You can change a device name using tools in the device's operating system.

SKILLS ADAPTABILITY

ACTIVITY

▼ IDENTIFYING DEVICES ON YOUR NETWORK

Identify the name, an IP address and the MAC address of a device on a network that you use.

COMPONENTS OF WIRED AND WIRELESS SYSTEMS

Wired and wireless systems can be made up of a variety of components.

CABLE

SUBJECT VOCABULARY

Mbit/s the amount of data that can be transferred per second, measured in Megabits (1 Mb = 1 million bits)
Gbit/s the amount of data that can be transferred per second, measured in Gigabits (1 Gb = 1,000,000,000 bits)

Cables are used to connect devices in a wired network. In homes and small businesses, Cat5e cables are used for Ethernet connections. These cables are able to transfer data at 10 **Mbit/s**, 100 Mbit/s or 1 **Gbit/s**. Cat5e cable connects devices through their NICs. The device's NIC allows the computer to exchange data with other networked computers and contains LEDs that signal network activity.

Cat6 cables can be used to transfer data at 10 Gbit/s. These are more expensive than Cat5e cables and are usually only used by businesses.



▲ **Figure 5.7** Cat5e cable is used to connect devices through their NICs

DID YOU KNOW?

400 Gbit/s Ethernet is expected to be available by late 2017.

DID YOU KNOW?

The speed of a network is limited by its slowest part. For example, if a network uses 10 Gbit/s cables and its internet connection speed is 100Mbit/s, then the internet connection is the slowest part of the network. This means that the 10 Gbit/s cables will not speed up the internet download speeds. The cables will only be transferring data at 1/100 of their maximum.

Fibre optic cables are flexible fibres. Each fibre optic cable contains a glass thread that bounces light signals between two devices faster and further than is possible with wire cables. Fibre optic cables can now carry data at 40 Gbit/s over many kilometres without affecting signal quality.

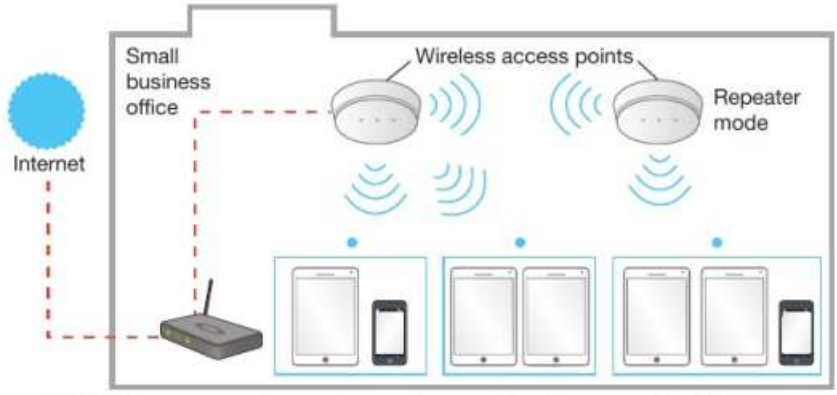
Fibre optic cables are expensive. This means that they are used by telecommunications companies and by organisations that need very fast data transfer speeds, such as science and engineering laboratories, hospitals, banks, schools and universities.



▲ **Figure 5.8** Fibre optic cable is capable of very high transfer speeds

WIRELESS ACCESS POINT

A wireless access point allows devices with Wi-Fi connectivity to connect to a wired network. They are often built into other hardware, such as routers (see page 85), but they are also available as stand-alone devices that connect to a wired network using Ethernet cables.



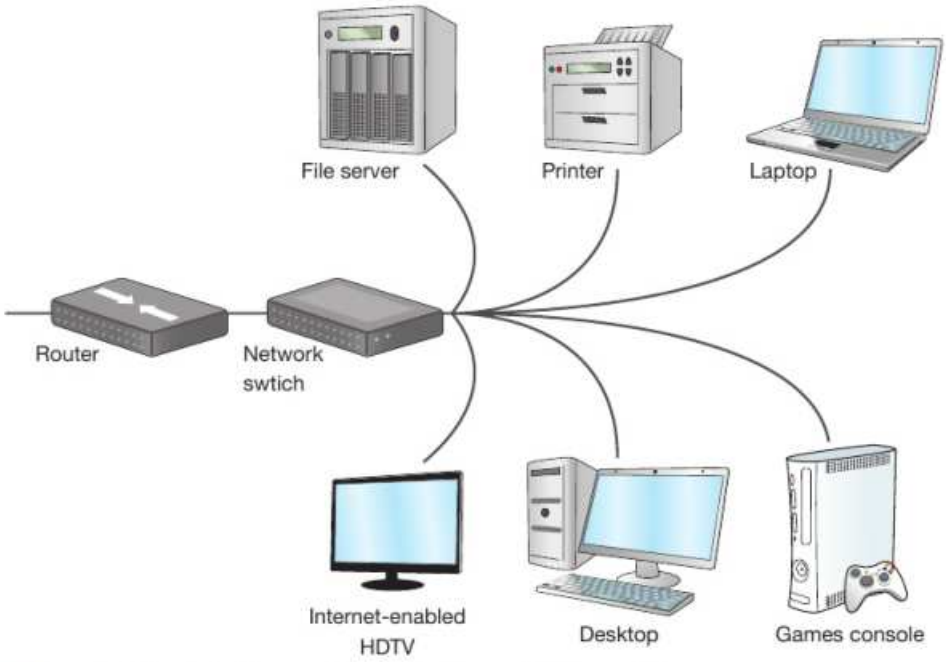
▲ **Figure 5.9** Wireless access points can be used in a small business to provide Wi-Fi access to wireless devices in a network

SWITCH

SUBJECT VOCABULARY

port a socket into which cables and devices can be plugged

A switch connects devices on a network. It has **ports**, each of which can be connected to a device using a cable. Connecting a wireless access point to a switch via a cable gives wireless devices access to the wired network. The switch makes sure that data sent from any device gets to the correct device on the network. For example, when printing a document, a laptop will send data that includes the printer's IP address to the switch and the switch then sends the data to the connected printer.



▲ **Figure 5.10** A switch allows multiple devices to send data to each other



▲ **Figure 5.11** A switch

GATEWAY

A gateway connects two different types of network. For example, a LAN is connected to a WAN using a gateway.

ROUTER

A router stores the addresses of all devices that are connected to it so that it can forward network traffic to its destination using the quickest route.

Most routers used in homes include a switch and a wireless access point. They also act as gateways, connecting the LAN to the internet, which is a type of WAN. Home routers are dynamically allocated an IP address by the internet service provider (ISP).



▲ **Figure 5.12** This router has a four-port switch; the last (grey) port on the right is used to connect the router to a WAN (in this case, the internet)

BOOSTER

A booster is used to amplify the signal in a network so that its range can be extended. For homes and offices, wired Ethernet connections often have a maximum range of 100 m. Wireless signals have limited range, too. Boosters can be used for both wired and wireless connections.

Wireless access points can be set to repeater mode in order to act as boosters for Wi-Fi signals, as shown in Figure 5.9.

SERVER

SUBJECT VOCABULARY

client a computer connected to a server

A server is a computer that shares its resources with connected devices. Computers connected to a server are known as **clients**. Resources that can be shared by one or more servers include printers, storage and applications.



▲ **Figure 5.13** Servers are often more powerful than clients, with lots of memory and fast processors to enable them to process multiple requests for their resources

SUBJECT VOCABULARY

Hypertext Transfer Protocol (HTTP) a set of standards that control how computer documents that are written in HTML connect to each other

AUTHENTICATION SERVER

An authentication server checks usernames and passwords. When a user successfully logs in, the client receives an electronic certificate that it can then use to access various resources, including applications and storage.

PRINT SERVER

A print server manages multiple printers at a time, dealing with print requests from client computers and adding jobs to a queue so that individual printers are not overloaded with requests. Print servers can also monitor and process print requests, making sure that users or departments can be invoiced for the jobs that they send to the printers.

FILE SERVER

File servers allow users to access shared and private storage.

APPLICATION SERVER

Application servers provide clients with access to applications that can be run directly from the server.

HINT

A single computer can be used to perform multiple server functions. Performing multiple roles will affect a computer's performance. Its ability to perform multiple roles will depend on its resources, especially its memory and processor.

WEB SERVER

Web servers process requests for data made via **Hypertext Transfer Protocol (HTTP)**. Together, all of the content stored on all web servers is known as the World Wide Web. Client computers often access web servers from outside the LAN to which the server is connected.

You will learn more about the difference between the internet and the World Wide Web in *Unit 3 Operating online* (page 110).

DID YOU KNOW?

HTTP was developed in 1989 by Sir Tim Berners-Lee as a way of sharing information with colleagues. He is credited with inventing the World Wide Web.

CONNECTING TO AND USING THE INTERNET

In order to access the online services provided by servers and data centres, users must have a connection to the internet. Users also need software that allows them to use and work with the services effectively and safely.

INTERNET SERVICE PROVIDER (ISP)

SUBJECT VOCABULARY

telecommunications infrastructure the networks of hardware facilities, owned by private and public organisations, that are used to transfer data

To connect to the internet, users need to subscribe to an ISP. ISPs provide connections to the **telecommunications infrastructure** that forms the framework for the internet. ISPs provide access via mobile telephone networks and landline telephone networks. Commercial ISPs charge subscription fees for access to the internet. Some ISPs provide free access as part of community schemes, which aim to provide internet access to groups of residents who either do not have or do not want access to commercial ISPs.

WEB BROWSER

A web browser is a type of software application used to request and display information stored on web servers. Examples of web browsers are Mozilla Firefox, Google Chrome, Internet Explorer® or Microsoft Edge, Opera[8], and Safari®[9].



▶ **Figure 5.14** Web browsers have versions available for mobile devices

SEARCH ENGINE

A search engine provides users with a way to find information in web pages stored on web servers. Users enter keywords that describe the information they want to find. The search engine then compares the keywords with those in its database of web pages and returns the results that are the closest match to the given keywords.



▲ **Figure 5.15** A search engine

8 OPERA IS A TRADEMARK OF OPERA SOFTWARE A

9 SAFARI® IS A TRADEMARK OF APPLE INC., REGISTERED IN THE U.S. AND OTHER COUNTRIES

HINT

Many people confuse search engines with web browsers. A search engine allows users to search and find information using keywords. A web browser is an application that users can use to view online content. Users can use a web browser to access a search engine.

DID YOU KNOW?

Some search engines allow users to search for information by uploading an image that represents the information they are looking for. The web browser then compares the image against a database of pages that include similar images and returns the results that are the closest matches.

You will learn more about search engines in *Unit 3 Operating online* (page 160–163).

FILTER SOFTWARE

SUBJECT VOCABULARY

URL (uniform resource locator) a website address
blacklist a list of unacceptable URLs
whitelist a list of acceptable URLs

Filter software prevents users from accessing inappropriate information. When a user tries to access a web page, the address (**URL**) and/or the contents of the web page are compared against two lists of URLs and keywords stored in the filter software's database. The two lists are the **blacklist** and the **whitelist**.

- If the results match anything in the blacklist, the user will be prevented from viewing the web page.
- If the result matches anything in the whitelist, then the user will be allowed to view the web page.
- If the result does not match anything in either the blacklist or the whitelist, the user will be allowed to view the information.

Administrators can add URLs to the blacklist and whitelist. The blacklist can be updated during software updates.

Filter software can help schools and parents to protect children from accessing disturbing or age-inappropriate content.

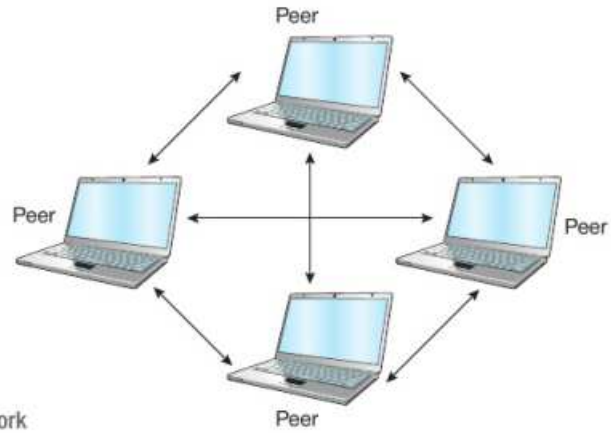
LOCAL AREA NETWORKS (LANS)

A LAN is a network contained to a small area, such as a home or office network (see page 69 for more information about LANs). Computers in a network can be connected using one of two different models:

- peer-to-peer
- client-server.

PEER-TO-PEER NETWORKS

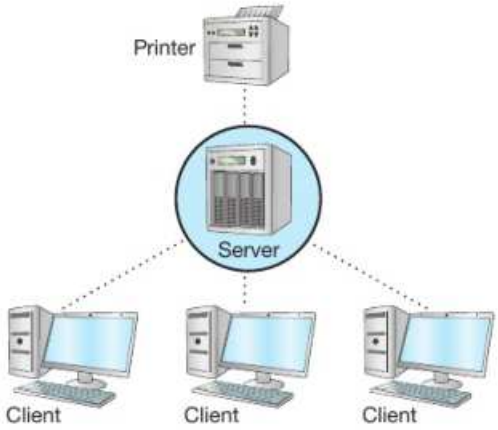
Computers in a peer-to-peer network share their resources with other computers in the network, but they do not access servers. Figure 5.16 is an example of a peer-to-peer network.



▶ **Figure 5.16** A peer-to-peer network

CLIENT-SERVER NETWORKS

Some networks use servers (see page 86). A network that uses servers and clients is called a client-server network. Figure 5.17 is an example of a client-server network.



▲ **Figure 5.17** A client-server network

BENEFITS OF USING LANS

Connecting computers using a LAN provides a range of benefits. These include:

- access to shared peripherals
- access to shared storage and data
- flexible access (that is, being able to access peripherals, storage and data from any connected device)
- media streaming (including movies, music and gaming)
- communication (that is, being able to send messages and files to others on the network)
- shared access to the internet.

SKILLS INTELLECTUAL INTEREST AND CURIOSITY
ADAPTABILITY
SELF-DIRECTION

ACTIVITY

▼ LAN PARTIES

What is a 'LAN party'? Use the internet to find out and write your own definition of the term.

BENEFITS OF USING CLIENT-SERVER NETWORKS

There are several benefits of using client-server networks that are not available when using a peer-to-peer network.

- **Control of user access rights:** Users, or groups of users, can be given access to some resources (such as storage or printers) and restricted from accessing others.
- **Centralised administration:** Resources and user accounts can be managed by an individual, or individual group of servers and administrators. This ensures that support can be provided by people who have an overview of the network and avoids inexperienced users creating problems for themselves or others.
- **Centralised backup:** User data is protected from loss because backups can be automated for all users. This makes it more likely that backups will happen than if individual users were asked to complete backups themselves.

- **Shared software:** Application servers (see page 86) can provide access to shared software. Some servers can provide access to operating systems.
- **Shared storage and file access:** The amount of storage available to users can be managed centrally. Sharing storage means that users can make files available to others. File permissions can be set for individual files, folders or drives, allowing users to either read only or read and write to different files.
- **Roaming profiles:** This is the ability to log into any computer in an office and see your settings and files. This allows users to access data, applications, mail and printers from any client, enabling them to work from anywhere where there is a client.

HINT

You learned about sharing software, storage and file access in *Unit 1 Digital devices* (pages 4–63).

SECURING DATA ON NETWORKS

Security prevents unauthorised users from accessing network resources and data.

LOGINS AND PASSWORDS

Users log in to computers on a network to access centrally managed resources. Without the correct **login** details, users cannot access the network or its resources.

Passwords are used to **authenticate** a user to the network. An authentication server can be used to manage the authentication of a user to a range of network resource and services. This means that, once a user has been authenticated, they do not have to log in each time they access a resource or service.

See page 86 for more information about authentication servers. You will learn more about passwords in *Unit 3 Operating online* (page 100).

SUBJECT VOCABULARY

login user or account information, such as a user name or account name
authenticate confirm that the user is who they say they are

FIREWALLS

A firewall is used at the gateway to a network. It controls the network traffic to and from a network, particularly the traffic from the internet. Firewalls prevent unauthorised users from accessing network devices and resources, such as storage. Firewalls are available as hardware and software, which can be installed on computers to prevent attacks from within a network.

ENCRYPTION

Encryption is the process of encoding, scrambling or jumbling data so that unauthorised users are prevented from being able to understand it.

SKILLS INTELLECTUAL INTEREST AND CURIOSITY
CREATIVITY
ADAPTABILITY
SELF-DIRECTION
COMMUNICATION

ACTIVITY

▼ ANCIENT ENCRYPTION

Research how a tool called a scytale was used by Spartans to encrypt messages. Create your own scytale and then use it to produce an encrypted message.

SUBJECT VOCABULARY

cipher a code

One method used to encrypt text is called a Caesar **cipher**. This method shifts each letter to the left by a set number of places. The number of places by which the letters have been shifted is known as the key.

Figure 5.18 shows an example of using a Caesar cipher. The top row is the original alphabet. The bottom row shows the alphabet after applying a Caesar cipher with a shift of four places. The example shows how the original letter F is encrypted to B.